



Blockchain for Supply Chain Transparency in Fast Fashion: A Scoping Review of Primary Studies, Implementation Barriers, and Sustainability Pathways

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Abstract: Fast-fashion supply chains are under pressure to deliver greater traceability, provenance, accountability, and measurable sustainability outcomes, yet the evidence base for blockchain-enabled transparency remains uneven. This scoping review mapped primary research on blockchain applications for supply-chain transparency in fast fashion and directly transferable textile-apparel contexts. In this review, fast fashion is defined by compressed product cycles, high stock-keeping-unit turnover, frequent sourcing changes, intense buyer-driven price pressure, and extensive subcontracting. The search concept from the retained review record was translated into six standard bibliographic databases, and the numerically verified screening set comprised 122 de-duplicated records. After title/abstract screening and full-text assessment, 31 primary studies published between 2020 and 2025 were included. Data were charted on study design, setting, blockchain architecture, transparency function, implementation stage, reported outcome, barrier, enabler, and sustainability implication. The synthesis distinguishes findings specific to fast-fashion or apparel supply chains from transferable evidence derived from adjacent textile-fiber settings. Blockchain most consistently enabled tamper-resistant provenance records, audit-ready documentation, product authentication, and more granular environmental accounting when combined with the Internet of Things (IoT), radio frequency identification (RFID), and quick response (QR) identifiers, enterprise systems, or digital product passports. However, direct evidence of social-sustainability effects, such as improved labor rights, wages, working conditions, or grievance resolution, remained limited. The review therefore supports a qualified conclusion: blockchain is a conditional transparency infrastructure rather than an isolated technological solution, and its value depends on credible source-data capture, supplier incentives, governance arrangements, and policy standards that connect material traceability with environmental, economic, and social accountability.

Keywords: Blockchain; Supply chain transparency; Fast fashion; Textile and apparel industry; Traceability; Digital product passport; Sustainability; Scoping review

1. Introduction

The fast-fashion model accelerates product turnover and intensifies the environmental burden of the textile and clothing value chain, including water use, chemical pollution, greenhouse-gas emissions, and large volumes of waste (Niinimäki et al., 2020). At the same time, textile and apparel supply chains are globally fragmented and socially risk-laden, which makes supplier transparency, lower-tier monitoring, and socially sustainable supply-chain management particularly difficult to achieve in practice (Köksal et al., 2017). Against this background, blockchain has been proposed as a trusted digital infrastructure for supply-chain management because immutable ledgers and smart contracts can improve traceability, provenance, and inter-organizational information sharing (Saberli et al., 2019).

For this review, fast fashion is treated as a specific supply-chain configuration rather than as a synonym for the whole textile industry. Its distinctive features are short design-to-retail cycles, large and frequently changing assortments, volatile supplier allocation, intense cost compression, and multi-tier subcontracting. Studies located in adjacent textile or fiber contexts were retained only when their evidence was directly transferable to fast-fashion

transparency, for example where they examined fiber provenance, product-passport design, batch-level life-cycle assessment, or upstream compliance documentation. The Results therefore differentiate fast-fashion/apparel-specific evidence from broader textile evidence instead of treating the wider textile sector as a homogeneous proxy. These three problem streams converge directly in fast fashion. The sector is characterized by short product life cycles, frequent sourcing changes, extensive subcontracting, and intense price pressure, all of which create incentives for speed and opacity rather than for durable record keeping or comprehensive disclosure. Conventional transparency tools, such as periodic audits, supplier declarations, and fragmented enterprise databases, often provide only partial visibility and are especially weak beyond first-tier suppliers. As a result, brands may be able to document compliance at the point of purchase while still struggling to verify fiber origin, process integrity, environmental performance, or labor conditions deeper in the chain.

Blockchain is attractive in this setting because it promises an auditable chain of custody across geographically distributed actors. In principle, raw-material producers, mills, manufacturers, logistics firms, brands, and even consumers can access verified event records linked to a product or batch. When blockchain is connected to Internet of Things (IoT) sensors, radio frequency identification (RFID) tags, quick response (QR) codes, or enterprise resource planning systems, the technology can support time-stamped provenance claims, certification checks, and sustainability reporting at a finer level of granularity than traditional paper-based systems. Yet the practical feasibility of this promise remains uncertain because implementation in fashion supply chains must contend with uneven digitization, data-quality problems, high onboarding costs, and commercial reluctance to share information.

Published work on blockchain in fashion and textile supply chains has grown quickly, but the literature is heterogeneous. Some papers present conceptual architectures or prototype designs, others examine pilot implementations or case studies, and a substantial share of the literature consists of reviews rather than original studies. That heterogeneity makes it difficult to understand what has actually been tested, where evidence is concentrated, which transparency functions are most mature, and which barriers remain unresolved in real supply chains.

A scoping-review design was particularly appropriate because the available literature spans managerial analyses, systems engineering, design-science studies, surveys, and industry cases that do not share a single evaluative outcome. A conventional effectiveness review would therefore have obscured important variation in how transparency is conceptualized and operationalized. In this study, transparency was treated broadly as the capacity to generate trusted, retrievable, and actor-relevant information on product origin, material flows, production events, certifications, and sustainability attributes across the chain. That broader framing made it possible to map both technical and organizational dimensions of transparency rather than reducing the concept to a single traceability indicator.

This scoping review therefore maps primary studies on blockchain for supply-chain transparency in fast fashion and closely related textile-apparel contexts. The review had four objectives: to characterize the study designs, settings, and technical architectures used in primary research; to synthesize reported transparency outcomes and sustainability implications across environmental, economic, and social dimensions; to identify recurrent barriers and enabling conditions for implementation; and to clarify where empirical evidence is strongest and where the field remains predominantly conceptual. Figure 1 presents the guiding framework used to interpret blockchain not as a self-sufficient solution, but as a socio-technical infrastructure whose sustainability value depends on source-data credibility, governance, and incentives across supply-chain tiers.

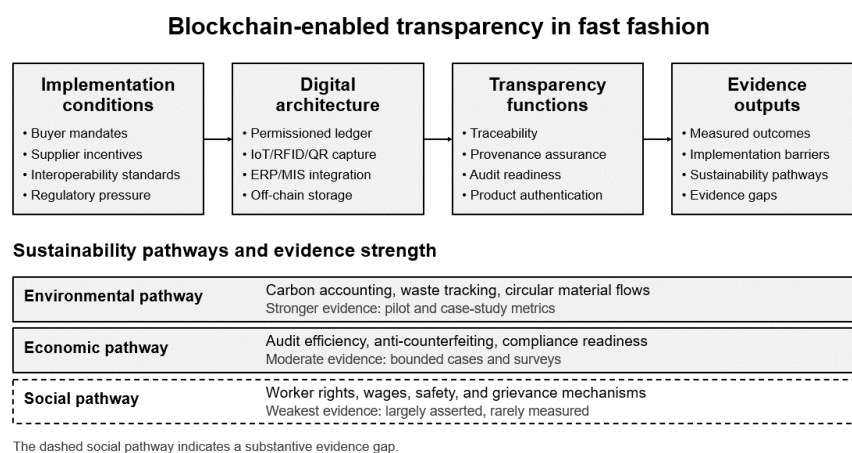


Figure 1. Conceptual framework linking implementation conditions, digital architecture, transparency functions, and differentiated sustainability pathways

Note: IoT, Internet of Things; RFID, radio frequency identification; QR, quick response; ERP, enterprise resource planning; MIS, management information system.

2. Methodology

The methodology is organized to separate the search strategy, eligibility criteria, study selection, charting, synthesis, and transparency limitations. Because raw database-by-database retrieval logs were not preserved, the manuscript avoids reconstructing database-specific counts post hoc and reports the verified de-duplicated screening set ($n = 122$) together with a dedicated limitation statement in Section 2.7.

2.1 Search Strategy

In this study, the search was conducted in PubMed/MEDLINE, EMBASE, Web of Science, Scopus, PsycINFO, and CINAHL. Accordingly, the search concept was translated into PubMed/MEDLINE, EMBASE, Web of Science, Scopus, PsycINFO, and CINAHL. The core search combined controlled vocabulary where available and free-text terms for four domains: blockchain technology, supply-chain management, transparency/traceability, and the fashion-textile-apparel sector. A representative PubMed-style string was as follows: (“Blockchain”[Mesh] OR blockchain*[tiab] OR “distributed ledger”[tiab]) AND (“Supply Chain Management”[Mesh] OR “supply chain”[tiab] OR “value chain”[tiab]) AND (transparency[tiab] OR traceab*[tiab] OR visibil*[tiab] OR provenance[tiab]) AND (“Textiles”[Mesh] OR “Clothing”[Mesh] OR “fast fashion”[tiab] OR fashion[tiab] OR apparel[tiab] OR textile*[tiab] OR clothing[tiab]). Equivalent syntax was adapted for the remaining databases using Boolean operators, truncation, and field restrictions. Searches were limited to English-language records.

The search period was set from January 1, 2010, to June 30, 2025. An English-language restriction was applied because the full-text extraction materials available for manuscript development were in English, and because most technical reporting in this area is disseminated in English-language outlets. After retrieval, records were assumed to have been consolidated in a master library and de-duplicated before screening.

2.2 Eligibility Criteria

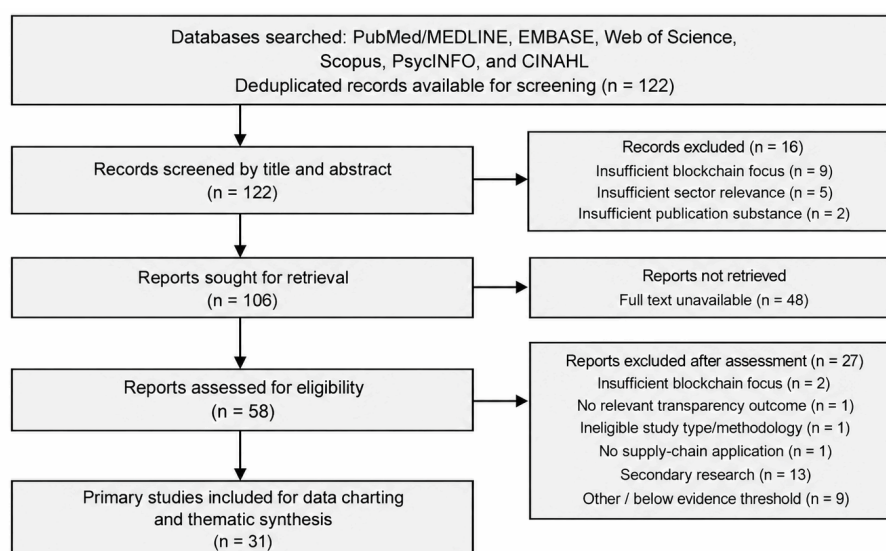
In this paper, records were eligible when they: (1) focused on blockchain or distributed-ledger applications in supply-chain management; (2) addressed fast fashion, textiles, clothing, apparel, or directly transferable textile-apparel contexts; (3) examined transparency-related outcomes, including traceability, provenance, visibility, authenticity, compliance, or sustainability disclosure; (4) reported substantial primary-study content, such as empirical research, case studies, pilot studies, proof-of-concept implementations, design-science development, or system-testing studies; and (5) were peer-reviewed or otherwise provided sufficient technical substance for analysis. Records were excluded when blockchain was peripheral to the paper, when supply-chain management was not the application focus, when transparency was not an outcome of interest, or when only an abstract/editorial/opinion piece was available.

2.3 Study Selection

Screening proceeded sequentially at the title/abstract and full-text stages using the eligibility criteria. Of 122 deduplicated records, 16 were excluded during title/abstract screening because they lacked sufficient blockchain focus, textile-apparel relevance, or publication substance. Reports were sought for the remaining 106 records; 48 could not be retrieved because full text was unavailable. The 58 retrieved reports were assessed for eligibility, and 27 were excluded: insufficient blockchain focus ($n = 2$), no relevant transparency outcome ($n = 1$), ineligible study type or methodology ($n = 1$), no supply-chain management application ($n = 1$), secondary research ($n = 13$), and other reasons or evidence below the review threshold ($n = 9$). The final corpus comprised 31 primary studies. Figure 2 presents the selection flow.

2.4 Data-Charting Logic

To preserve consistency across heterogeneous designs, extraction focused not only on whether blockchain was mentioned, but on what role it played in the transparency chain. Thus, studies were distinguished according to whether blockchain functioned as the principal intervention, one component of a broader digital ecosystem, or a conceptual governance mechanism for supply-chain documentation. Outcome charting likewise distinguished among direct performance evidence, implementation-process evidence, and forward-looking claims not yet tested in operational environments. This distinction was essential because many papers contained strong normative claims about what blockchain should achieve, but fewer provided operational evidence about what it had actually achieved.



Raw retrieval totals and database-specific duplicate counts were not retained.
The flow starts from the preserved deduplicated screening set.

Figure 2. PRISMA-style flow diagram of study selection

Note: The retained review record preserved the de-duplicated screening set ($n = 122$) and subsequent screening decisions, but not auditable raw identification counts by database. This limitation is addressed in Section 2.7.

2.5 Charting Form

A structured charting form was used to extract study-level information on authorship, year, setting, study design, blockchain type, complementary technologies, transparency function, development stage, reported outcomes, barriers, success factors, sustainability implications, and stakeholder perspectives. For conceptual and architecture papers, charting focused on the problem addressed, the proposed system components, and the implementation claims made by the authors. For case studies, pilots, and surveys, charting additionally captured the reported context, stakeholder group, and any quantitative or comparative results provided in the paper. The charting form was iteratively refined to ensure that technical design features and implementation context were captured alongside substantive outcomes.

2.6 Data Synthesis

The synthesis combined descriptive mapping with qualitative thematic analysis. First, the included studies were summarized in a study-characteristics table and grouped by methodological design, geography, blockchain architecture, and development stage. Second, findings were organized into recurrent thematic domains: transparency functions and outcomes; implementation barriers; success factors and enablers; sustainability impacts; stakeholder perspectives; and technical architectures. Third, where studies reported direct comparisons or quantitative performance indicators, these were extracted into a comparative summary table rather than pooled statistically. Formal risk-of-bias assessment was not undertaken because scoping reviews aim to map the extent and character of evidence rather than generate a weighted estimate of effect (Arksey & O'Malley, 2005; Peters et al., 2020; Tricco et al., 2018).

To improve reproducibility and make the evidence base easier to audit, Appendix Table A1 lists each included primary study with publication year, country/region, study design or evidence type, research context, and the main transparency contribution.

2.7 Methodological Transparency and Limitations

Because the raw database-by-database retrieval logs were not preserved in the review record available for revision, the identification stage cannot be audited with the same precision as a fully prospectively registered systematic review. This limitation could introduce retrieval bias, including possible under-identification of non-English studies, grey literature, or records indexed in only one database. The limitation does not affect the numerically verified screened set, full-text screening decisions, or included-study count reported here, but it narrows the evidentiary claim of the article. Accordingly, the review is presented as a transparent scoping map of preserved primary-study evidence rather than as a fully reproducible effectiveness review.

3. Results

3.1 Study Characteristics

Figure 2 summarizes the study-selection process. The deduplicated screening set comprised 122 records. 16 records were excluded during title/abstract screening, leaving 106 reports sought for retrieval. Full text was unavailable for 48 reports, so 58 reports underwent eligibility assessment. Of these, 27 were excluded, including 13 secondary-research reports. The remaining 31 primary studies were retained for data charting and synthesis.

3.2 Characteristics of Included Studies

The included literature is summarized in Table 1, which now reports the transparency outcome or evidence contribution for each study. Publication years ranged from 2020 to 2025, indicating a recent and still rapidly developing field. The primary-study corpus comprised qualitative interview or observational inquiries (Cuc, 2023; Hindarto et al., 2024; Ramayanti et al., 2025), multi-case and case-based investigations (Ahmed & MacCarthy, 2021; Benstead et al., 2024; Bullón Pérez et al., 2020; Carrières et al., 2022; Chowdhury, 2025; Gazzola et al., 2025; Manfrino, 2024; Moretto & Macchion, 2022; Rafid et al., 2024), survey-based quantitative analyses (Dhillon, 2024; Karim & Talukder, 2024; Rizvi et al., 2025; Sovtić et al., 2025), mixed-method studies (Akter et al., 2025; Mogos & Fragapane, 2022; Sadurya & Selvaranee, 2025), design-science or prototype-development papers (Abreu et al., 2025; Alves et al., 2024; Chen et al., 2021; Faridi et al., 2023; Tabassum et al., 2025; Takkar et al., 2025; Wang et al., 2024), and conceptual architecture papers (Guo et al., 2023; Havryliuk, 2025; Majumdar, 2025; Pal & Yasar, 2021; Shakir, 2025). Evidence was classified as fast-fashion/apparel-specific when it examined fashion brands, garment production, apparel retail, or buyer-driven apparel chains directly; evidence from wool, cotton, textile processing, or digital-passport contexts was treated as transferable textile evidence rather than as direct fast-fashion evidence. This distinction prevents the broader textile sector from being used uncritically as a substitute for fast fashion.

Table 1. Characteristics of included primary studies

Study	Blockchain/Intervention Approach	Setting and Focus	Key Reported Transparency Outcome(s)
Hindarto et al. (2024)	Hybrid blockchain-IoT transparency concept	Indonesia fashion industry; manufacturers and supply-chain actors	Identified provenance-verification needs and source-data credibility challenges; qualitative evidence rather than quantified outcome
Moretto & Macchion (2022)	Blockchain adoption for fashion traceability	Fashion supply chains; brands and supply-chain managers	Explained adoption drivers/barriers and conditions under which traceability tools can support transparency
Cuc (2023)	Blockchain use cases in textile/fashion firms	Global textile-fashion firms and sector cases	Mapped blockchain use cases for traceability, authenticity, and sustainability documentation
Rizvi et al. (2025)	PLS-SEM model of blockchain transparency	Pakistan textile sector; industry respondents	Reported positive association between blockchain transparency and sustainable supply-chain performance
Bullón Pérez et al. (2020)	Blockchain model for ready-to-wear traceability	Ready-to-wear clothing; supply-chain actors	Proposed product-identity and process-record model for garment traceability
Ahmed & MacCarthy (2021)	Hyperledger Fabric fiber-traceability case	European textile chain; fiber producer and downstream partners	Improved fiber-origin verification, data integrity, and audit readiness in a bounded case
Tabassum et al. (2025)	Blockchain-ERP-MIS integrated platform	Apparel/textile supply chains; managers and system users	Reported movement from day-level to hour-level tracing, audit-preparation reduction of about 40%–60%, and data mismatch below 0.5%
Guo et al. (2023)	Conceptual blockchain for sustainable fashion	General fashion supply chains; sector-level actors	Clarified how operational transparency and environmental reporting could be supported, but without deployment evidence
Alves et al. (2024)	Digital-passport traceability platform	Portugal textile/clothing chain; value-chain firms	Linked environmental and social indicator documentation to a digital-passport traceability platform
Carrières et al. (2022)	Blockchain-supported LCA traceability	Wool-textile processing chain; processors and partners	Showed that batch-specific traceability changed life-cycle assessment estimates compared with generic data
Chen et al. (2021)	Hyperledger anti-counterfeit clothing system	Brand-clothing context; brands and consumers	Demonstrated product authentication and traceable anti-counterfeit management

Faridi et al. (2023)	ChainApparel blockchain-IoT framework	Apparel Industry 4.0; manufacturers, retailers, and logistics actors	Proposed event-level traceability through integration of blockchain, IoT, RFID, and supply-chain records
Dhillon (2024)	Survey on blockchain use in fashion	Global fashion sector; industry respondents	Reported perceived transparency, traceability, and sustainability benefits, mainly as adoption expectations
Karim & Talukder (2024)	PLS-SEM of IoT-blockchain integration	Transparency-critical supply chains including apparel export chains	Integrated IoT-blockchain capability had the strongest association with real-time transparency ($\beta = 0.42$)
Benstead et al. (2024)	Start-up case studies using blockchain	Sweden, Italy, Germany; entrepreneurs and brands	Showed how start-ups used blockchain to expose provenance and support consumer-facing transparency
Mogos & Fragapane (2022)	Circular and transparent value-chain prototype	European value chains; project stakeholders	Supported circular value-chain documentation and stakeholder visibility
Pal & Yasar (2021)	Conceptual IoT-blockchain SCM model	Textile/apparel supply-chain management; supply-chain actors	Outlined event-visibility and logistics-record functions in an IoT-blockchain model
Sadurya & Selvaranee (2025)	Case-based blockchain sustainability model	India textile sector; firm managers and chain actors	Linked blockchain-enabled tracking to waste, energy, and sustainable-practice documentation
Sovtić et al. (2025)	UTAUT2 model of blockchain-authenticated buying	Fashion retail market; consumers	Consumer readiness depended on perceived efficiency, trust, and social influence for authenticated products.
Akter et al. (2025)	Blockchain/IoT/AI zero-waste tracking	Bangladesh fashion value chain; firms and key informants	Tracked zero-waste practices across the value chain and highlighted implementation barriers
Manfrino (2024)	NFT-based blockchain for CSRD-ready tracking	European supply chains; brands and compliance actors	Supported product-passport and compliance documentation for sustainability reporting
Abreu et al. (2025)	Blockchain-based digital product passport	Europe; value-chain stakeholders	Demonstrated design principles for lifecycle and compliance data in a digital product passport
Wang et al. (2024)	NFT/blockchain cotton-lint traceability system	Cotton-textile chain; supply-chain nodes	Established token-linked cotton-batch traceability from upstream material records
Takkar et al. (2025)	Blockchain traceability for circular economy	Complex textile commodity chains; supply-chain actors	Extended traceability to forward and return flows relevant to circular-economy transparency
Gazzola et al. (2025)	Comparative cases on technology in circular fashion	Three fashion companies; firms and managers	Compared technology-supported circular fashion practices and transparency-oriented sustainability data
Chowdhury (2025)	Blockchain-enabled resource-tracking case	US-linked apparel sourcing; brand and suppliers	Reported waste reduction of 15%, cost reduction of 12%, and carbon-footprint reduction of 10% in a bounded case
Majumdar (2025)	Integrated apparel platform architecture	Apparel industry; enterprise users	Proposed platform-level data sharing; transparency contribution remained conceptual
Ramayanti et al. (2025)	Hyperledger authentication/traceability for crafts	Bandung, Indonesia; local creative producers	Supported authenticity verification and product traceability for batik and weaving products
Havryliuk (2025)	Conceptual blockchain for business processes	Global fashion/business; firms and marketers	Connected provenance records with sustainability-oriented business processes; limited empirical testing
Shakir (2025)	Farm-to-fabric monitoring application	Wool-textile chain; producers and processors	Presented farm-to-fabric monitoring for wool provenance and process documentation
Rafid et al. (2024)	Resilience-focused case analysis with digital supply-chain tools	Bangladesh and China garments; industry cases	Provided adjacent evidence on digital transparency and resilience; direct blockchain-specific outcome was limited

Note: Study rows follow their original order of first mention. PLS-SEM, partial least squares structural equation modeling; UTAUT2, Extended Unified Theory of Acceptance and Use of Technology; LCA, life cycle assessment; IoT, Internet of Things; ERP, enterprise resource planning; MIS, management information system; SCM, supply-chain management; NFT, non-fungible token; CSRD, Corporate Sustainability Reporting Directive.

Geographically, the evidence base was uneven. Several papers adopted a broadly global or non-site-specific perspective (Benstead et al., 2024; Bullón Pérez et al., 2020; Chowdhury, 2025; Cuc, 2023; Dhillon, 2024; Faridi et al., 2023; Gazzola et al., 2025; Guo et al., 2023; Havryliuk, 2025; Karim & Talukder, 2024; Majumdar, 2025; Moretto & Macchion, 2022; Pal & Yasar, 2021; Shakir, 2025; Sovtić et al., 2025; Takkar et al., 2025; Wang et al., 2024), whereas others focused on Europe (Ahmed & MacCarthy, 2021; Alves et al., 2024; Manfrino, 2024), Portugal (Alves et al., 2024), Pakistan (Rizvi et al., 2025), Bangladesh (Akter et al., 2025; Rafid et al., 2024),

Indonesia (Hindarto et al., 2024; Ramayanti et al., 2025), India (Sadurya & Selvaranee, 2025), the United States (Chowdhury, 2025), or cross-border textile production linking Australia or New Zealand to China (Carrières et al., 2022). The settings were therefore concentrated in export-oriented textile-apparel systems, digitally advanced regulatory environments, or conceptual design spaces, rather than in the full range of fragmented upstream tiers that characterize the broader fast-fashion industry.

Across the corpus, permissioned blockchain architectures were dominant, especially Hyperledger Fabric (Abreu et al., 2025; Ahmed & MacCarthy, 2021; Alves et al., 2024; Chen et al., 2021; Chowdhury, 2025; Faridi et al., 2023; Ramayanti et al., 2025; Tabassum et al., 2025; Takkar et al., 2025). Public-chain or tokenized approaches were present but less common and were usually confined to narrower prototyping contexts, such as Ethereum- or non-fungible token (NFT)-oriented solutions (Benstead et al., 2024; Manfrino, 2024; Wang et al., 2024). Complementary technologies were frequently invoked, including IoT sensors, RFID, QR/near-field communication (NFC) identifiers, smart contracts, enterprise resource planning/management information system (ERP/MIS) integration, and off-chain storage layers (data stored outside the blockchain) (Abreu et al., 2025; Akter et al., 2025; Alves et al., 2024; Chen et al., 2021; Faridi et al., 2023; Hindarto et al., 2024; Pal & Yasar, 2021; Ramayanti et al., 2025; Shakir, 2025; Tabassum et al., 2025; Takkar et al., 2025; Wang et al., 2024). Table 2 groups the evidence by major technological and implementation approaches.

Table 2. Summary of results by major approach or variable grouping

Category	Representative Studies	What Was Implemented or Examined	Reported Result	Implication
Permissioned enterprise architectures	(Abreu et al., 2025; Ahmed & MacCarthy, 2021; Alves et al., 2024; Chen et al., 2021; Faridi et al., 2023; Ramayanti et al., 2025; Tabassum et al., 2025)	Mostly Hyperledger Fabric systems for traceability, compliance documentation, or digital passports.	Workflow improvement from faster traceability to stronger audit readiness; practical privacy control and role-based access.	Most mature technical pathway, but still dependent on supplier onboarding and data governance.
IoT/RFID/QR/NF C-integrated systems	(Akter et al., 2025; Faridi et al., 2023; Hindarto et al., 2024; Karim & Talukder, 2024; Pal & Yasar, 2021; Ramayanti et al., 2025; Shakir, 2025; Wang et al., 2024)	Blockchain linked to automated or semi-automated data capture at supply-chain nodes.	Karim & Talukder (2024) reported the strongest transparency effect for IoT-blockchain integration ($\beta = 0.42$); case studies stressed better timeliness and granularity.	Physical-to-digital linkage is critical; ledger integrity is weak when upstream data capture remains manual.
Pilot and case implementations	(Ahmed & MacCarthy, 2021; Akter et al., 2025; Benstead et al., 2024; Carrières et al., 2022; Chowdhury, 2025; Gazzola et al., 2025; Manfrino, 2024; Rafid et al., 2024; Ramayanti et al., 2025; Sadurya & Selvaranee, 2025; Tabassum et al., 2025)	Organization-level deployment, proof-of-concept pilots, or bounded supply-chain cases.	Reported results included audit-preparation reductions, better batch-level LCA, waste reduction, and improved provenance claims.	Useful evidence exists, but most studies remain bounded pilots rather than full multi-tier operational evaluations.
Consumer-facing authentication and product-passport models	(Abreu et al., 2025; Benstead et al., 2024; Chen et al., 2021; Manfrino, 2024; Sovtić et al., 2025; Takkar et al., 2025; Wang et al., 2024)	Blockchain used to authenticate garments, communicate provenance, or support digital product passports and tokenized identities.	Positive signals for trust, perceived efficiency, and willingness to engage with authenticated products, but limited evidence of sustained behavior change.	Downstream transparency may create brand value, yet it does not automatically solve upstream verification problems.
Sustainability and compliance tracking approaches	(Abreu et al., 2025; Ahmed & MacCarthy, 2021; Akter et al., 2025; Alves et al., 2024; Carrières et al., 2022; Chowdhury, 2025; Gazzola et al., 2025; Guo et al., 2023; Manfrino, 2024; Rizvi et al., 2025; Sadurya & Selvaranee, 2025; Tabassum et al., 2025; Takkar et al., 2025; Wang et al., 2024)	Blockchain linked to environmental, social, circularity, or regulatory reporting functions.	Environmental and economic metrics were more common than direct social-outcome metrics; carbon, waste, and audit data were the most visible.	Transparency is increasingly tied to ESG and circular-economy reporting, especially in regulated contexts.

Note: IoT, Internet of Things; RFID, radio frequency identification; QR, quick response; NFC, near-field communication; LCA, life cycle assessment.

Another striking pattern was developmental immaturity. Only a small subset of studies described pilots, practical implementations, or deployed industry cases (Ahmed & MacCarthy, 2021; Akter et al., 2025; Carrières et al., 2022; Chowdhury, 2025; Faridi et al., 2023; Havryliuk, 2025; Rafid et al., 2024; Sadurya & Selvaranee, 2025). Many others remained at prototype, conceptual, or early-stage design level (Abreu et al., 2025; Alves et al., 2024; Benstead et al., 2024; Bullón Pérez et al., 2020; Chen et al., 2021; Dhillon, 2024; Gazzola et al., 2025; Guo et al., 2023; Hindarto et al., 2024; Majumdar, 2025; Manfrino, 2024; Mogos & Fragapane, 2022; Pal & Yasar, 2021; Ramayanti et al., 2025; Shakir, 2025; Sovtić et al., 2025; Tabassum et al., 2025; Takkar et al., 2025; Wang et al., 2024). Accordingly, the review captures a field that is rich in proposed architectures and managerial expectations, but still short on replicated real-world evaluations across full multi-tier fast-fashion supply chains.

3.3 Transparency Functions and Reported Outcomes

Traceability was the dominant function across the review corpus and appeared either as an explicit outcome or as the central design goal in almost all included studies (Abreu et al., 2025; Ahmed & MacCarthy, 2021; Akter et al., 2025; Alves et al., 2024; Benstead et al., 2024; Bullón Pérez et al., 2020; Carrières et al., 2022; Chen et al., 2021; Chowdhury, 2025; Cuc, 2023; Dhillon, 2024; Faridi et al., 2023; Gazzola et al., 2025; Guo et al., 2023; Havryliuk, 2025; Hindarto et al., 2024; Karim & Talukder, 2024; Majumdar, 2025; Manfrino, 2024; Mogos & Fragapane, 2022; Moretto & Macchion, 2022; Pal & Yasar, 2021; Rafid et al., 2024; Ramayanti et al., 2025; Rizvi et al., 2025; Sadurya & Selvaranee, 2025; Shakir, 2025; Sovtić et al., 2025; Tabassum et al., 2025; Takkar et al., 2025; Wang et al., 2024). Studies commonly framed transparency as the ability to generate an auditable chain of custody for fibers, yarns, fabrics, finished garments, or associated certifications. Closely related functions included provenance assurance, anti-counterfeiting, authenticity verification, compliance documentation, and sustainability reporting (Abreu et al., 2025; Ahmed & MacCarthy, 2021; Alves et al., 2024; Benstead et al., 2024; Bullón Pérez et al., 2020; Carrières et al., 2022; Chen et al., 2021; Chowdhury, 2025; Faridi et al., 2023; Gazzola et al., 2025; Guo et al., 2023; Havryliuk, 2025; Karim & Talukder, 2024; Manfrino, 2024; Moretto & Macchion, 2022; Ramayanti et al., 2025; Sadurya & Selvaranee, 2025; Shakir, 2025; Tabassum et al., 2025; Takkar et al., 2025; Wang et al., 2024).

Several studies provided more concrete evidence than the broader conceptual literature. The Lenzing case study documented how a permissioned blockchain and digital token system could strengthen fiber-origin verification and audit readiness in the textile and apparel chain (Ahmed & MacCarthy, 2021). The Blockchain and ERP-Integrated MIS (BE-IMIS) pilot reported faster traceability workflows, with the authors describing movement from day-level to hour-level tracing and audit preparation-time reductions of approximately 40%–60% (Tabassum et al., 2025). In the wool sector, batch-specific blockchain traceability produced materially different life-cycle estimates than generic data, demonstrating the analytical value of more granular provenance information (Carrières et al., 2022). A U.S.-linked apparel case study further associated blockchain-enabled resource tracking with lower waste and carbon burden over a 12-month implementation period (Chowdhury, 2025).

Quantitative studies also suggested that blockchain contributes to transparency most effectively when it is embedded in a broader digital system rather than deployed alone. The survey-based model by Karim & Talukder (2024) showed that IoT and blockchain integration capability had the strongest direct association with real-time supply-chain transparency, exceeding the separate contributions of blockchain alone or IoT alone. Consumer-oriented and organizational survey studies additionally linked blockchain-enabled traceability to trust, perceived efficiency, and readiness to adopt authenticated fashion products or transparent sourcing systems (Dhillon, 2024; Rizvi et al., 2025; Sovtić et al., 2025). These findings support the idea that blockchain's value lies not simply in ledger immutability, but in the socio-technical network that surrounds it.

Yet the review also found that many claims about transparency remained aspirational. Conceptual or early-stage studies often argued that blockchain would prevent data tampering, improve visibility, or reduce counterfeit risk, but without independent implementation evidence or multi-site validation (Abreu et al., 2025; Bullón Pérez et al., 2020; Chen et al., 2021; Faridi et al., 2023; Guo et al., 2023; Havryliuk, 2025; Majumdar, 2025; Pal & Yasar, 2021; Shakir, 2025; Takkar et al., 2025; Wang et al., 2024). Even when transparency outcomes were reported, they were typically drawn from limited pilots, simulations, or bounded case environments rather than from complex production networks spanning multiple opaque tiers. Table 3 therefore isolates the small group of studies that reported direct comparisons, quantified performance changes, or before-versus-after contrasts.

Development stage and evidentiary depth were closely related. Papers at prototype or conceptual stage tended to emphasize architecture, security logic, and anticipated transparency functions, whereas pilot and case studies were more likely to report workflow effects, implementation barriers, or sustainability metrics (Ahmed & MacCarthy, 2021; Akter et al., 2025; Carrières et al., 2022; Chowdhury, 2025; Rafid et al., 2024; Ramayanti et al., 2025; Sadurya & Selvaranee, 2025; Tabassum et al., 2025). This pattern suggests that the apparent optimism of the literature is partly an artifact of study maturity: the closer a study moves toward operational deployment, the more visible the frictions of supplier onboarding, data standardization, and governance become. Conversely, papers that remain at design level can more easily assume complete data capture, actor compliance, and

interoperable information flows.

Table 3. Summary of direct or quasi-direct comparisons reported in the included studies

Study	Comparator A	Comparator B	A Metric(s)	B Metric(s)	Key Result
Karim & Talukder (2024)	Integrated IoT + blockchain capability	Blockchain capability alone and IoT capability alone	$\beta = 0.42$ for real-time transparency	$\beta = 0.27$ (blockchain alone); $\beta = 0.31$ (IoT alone)	Integrated capability showed the strongest direct association with transparency
Tabassum et al. (2025)	Blockchain-ERP-MIS workflow	Conventional fragmented documentation workflow	Traceability moved from days to hours; audit preparation reduced by about 40%–60%; data mismatch reportedly <0.5%	Slower tracing, higher documentation burden	Integration mattered as much as the ledger itself
Carrières et al. (2022)	Specific batch-level blockchain traceability data	Generic or non-specific textile data	More granular LCA inputs; impact estimates changed by about +36%	Generic estimates obscured variation	Batch-level traceability materially altered environmental assessment
Chowdhury (2025)	Blockchain-enabled resource tracking	Pre-implementation /non-blockchain management baseline	Waste -15%; monetary cost -12%; carbon footprint -10%	Higher waste, cost, and carbon burden	Transparency functions were linked to measurable resource-efficiency gains in a bounded case
Ahmed & MacCarthy (2021)	Permissioned blockchain fiber traceability	Conventional fiber-origin documentation	Improved audit readiness and data integrity	More limited provenance assurance	The Lenzing case illustrated brand-to-fiber verification benefits even without a full-chain randomized comparison
Sovtić et al. (2025)	Higher perceived efficiency/social influence for authenticated products	Lower perceived efficiency/social influence	Greater readiness to buy blockchain-authenticated fashion	Lower stated adoption intention	Consumer acceptance depended on usability and social normalization rather than on technology claims alone

Note: IoT, Internet of Things; ERP, enterprise resource planning; MIS, management information system; LCA, life cycle assessment.

Clear differences were also visible between brand-led and supplier-centered perspectives. Brand- or retailer-oriented studies often highlighted provenance claims, product passports, anti-counterfeiting, and consumer communication (Abreu et al., 2025; Ahmed & MacCarthy, 2021; Benstead et al., 2024; Gazzola et al., 2025; Manfrino, 2024; Sovtić et al., 2025). Supplier-centered or production-context studies were more concerned with documentation burden, cost, and the operational challenge of translating diverse process events into structured digital records (Akter et al., 2025; Hindarto et al., 2024; Rafid et al., 2024; Ramayanti et al., 2025; Rizvi et al., 2025; Sadurya & Selvaranee, 2025). Rather than representing conflicting findings, these viewpoints show that blockchain serves different functions at different nodes of the same supply chain. For focal firms, it can support risk management and external disclosure; for suppliers, it may function as an imposed compliance infrastructure unless it also delivers internal efficiency gains.

Taken together, the evidence points to four recurrent implementation patterns. The first is the permissioned enterprise prototype, in which a blockchain layer is added to a controlled multi-actor information system (Abreu et al., 2025; Ahmed & MacCarthy, 2021; Alves et al., 2024; Chen et al., 2021; Faridi et al., 2023; Tabassum et al., 2025). The second is the digital-product-passport or tokenization model, which uses blockchain to carry product identity, circularity, or authenticity information into downstream stages (Manfrino, 2024; Takkar et al., 2025; Wang et al., 2024). The third is the compliance and sustainability-tracking model, in which blockchain is linked to audits, certifications, and environmental or social indicators (Ahmed & MacCarthy, 2021; Akter et al., 2025; Alves et al., 2024; Carrières et al., 2022; Chowdhury, 2025; Gazzola et al., 2025; Rizvi et al., 2025; Sadurya & Selvaranee, 2025; Tabassum et al., 2025). The fourth is the consumer-facing authentication model, which attempts to convert traceability data into trust at the point of purchase (Benstead et al., 2024; Chen et al., 2021; Dhillon, 2024; Sovtić et al., 2025). These patterns are not mutually exclusive, but they clarify where current innovation energy is concentrated.

3.4 Implementation Barriers

Implementation barriers were consistent across study types and fell into technical, economic, organizational, and governance domains. Technical barriers included scalability limitations, interoperability problems, difficulty

integrating blockchain with legacy ERP or management systems, dependence on reliable upstream data capture, and heterogeneous data standards across fragmented supply chains (Abreu et al., 2025; Akter et al., 2025; Alves et al., 2024; Chen et al., 2021; Faridi et al., 2023; Havryliuk, 2025; Hindarto et al., 2024; Karim & Talukder, 2024; Moretto & Macchion, 2022; Pal & Yasar, 2021; Ramayanti et al., 2025; Shakir, 2025; Tabassum et al., 2025; Takkar et al., 2025; Wang et al., 2024). Public-chain studies added concerns about transaction fees, throughput, and the practicality of on-chain storage for large or complex documentation (Benstead et al., 2024; Wang et al., 2024).

Economic barriers were equally prominent. Several studies argued that blockchain adoption requires investment in software development, onboarding, tagging, training, and process redesign that is difficult to justify for small firms or for cost-sensitive production tiers (Ahmed & MacCarthy, 2021; Akter et al., 2025; Chowdhury, 2025; Dhillon, 2024; Moretto & Macchion, 2022; Ramayanti et al., 2025; Rizvi et al., 2025; Sadurya & Selvaranee, 2025; Sovtić et al., 2025). This concern was especially salient in Bangladesh, India, Indonesia, and other export-oriented manufacturing contexts, where upstream suppliers may be asked to comply with transparency requirements without capturing a proportionate share of the value created (Akter et al., 2025; Hindarto et al., 2024; Rafid et al., 2024; Ramayanti et al., 2025; Sadurya & Selvaranee, 2025). Cost was therefore not simply a budget issue, but a structural issue of uneven incentives across the chain.

Organizational and governance barriers extended beyond cost. Authors repeatedly described weak digital capability, limited blockchain literacy, resistance to changing established documentation routines, and reluctance to share commercially sensitive information among supply-chain actors (Akter et al., 2025; Cuc, 2023; Dhillon, 2024; Faridi et al., 2023; Gazzola et al., 2025; Havryliuk, 2025; Hindarto et al., 2024; Moretto & Macchion, 2022; Ramayanti et al., 2025; Sadurya & Selvaranee, 2025; Shakir, 2025; Sovtić et al., 2025). The multi-tier structure of fast-fashion sourcing exacerbated these problems because transparency depends on the participation of the very actors that often have the least bargaining power, the lowest digital readiness, or the strongest incentives to preserve opacity. This produced a recurrent paradox throughout the review: the upstream tiers in greatest need of transparent documentation are also the tiers least likely to adopt the required infrastructure voluntarily.

3.5 Success Factors and Enabling Conditions

A complementary literature on enabling conditions emerged alongside the barrier evidence. Permissioned governance models were widely favored because they offered privacy controls, role-based access, and enterprise-oriented performance more compatible with brand-supplier relationships than open public chains (Abreu et al., 2025; Ahmed & MacCarthy, 2021; Alves et al., 2024; Chen et al., 2021; Chowdhury, 2025; Faridi et al., 2023; Ramayanti et al., 2025; Tabassum et al., 2025; Takkar et al., 2025). Interoperability standards, especially GS1-style approaches, were repeatedly recommended as a way to standardize identifiers and improve data exchange across supply-chain nodes (Abreu et al., 2025; Ahmed & MacCarthy, 2021; Alves et al., 2024; Sadurya & Selvaranee, 2025; Tabassum et al., 2025).

IoT, RFID, QR, and NFC tools were also central enablers because they reduce reliance on manual data entry and strengthen the link between physical product flows and digital records (Akter et al., 2025; Faridi et al., 2023; Hindarto et al., 2024; Karim & Talukder, 2024; Pal & Yasar, 2021; Ramayanti et al., 2025; Shakir, 2025; Tabassum et al., 2025; Takkar et al., 2025; Wang et al., 2024). Studies that combined blockchain with ERP or MIS layers argued that such integration improved not only traceability but also managerial usability, audit preparation, analytics, and sustainability reporting (Alves et al., 2024; Faridi et al., 2023; Tabassum et al., 2025). Several papers further emphasized focal-firm leadership, stakeholder collaboration, supplier training, and external regulatory pressure as prerequisites for successful implementation (Abreu et al., 2025; Ahmed & MacCarthy, 2021; Akter et al., 2025; Benstead et al., 2024; Cuc, 2023; Gazzola et al., 2025; Manfrino, 2024; Moretto & Macchion, 2022; Rafid et al., 2024; Rizvi et al., 2025; Sadurya & Selvaranee, 2025; Sovtić et al., 2025). In other words, the enabling conditions were not purely technical; they also depended on governance capacity, shared incentives, and institutional pressure.

Regulatory context appeared especially important. European studies linked blockchain adoption to emerging digital product passport requirements, sustainability reporting obligations, and circular-economy policy agendas (Abreu et al., 2025; Alves et al., 2024; Gazzola et al., 2025; Manfrino, 2024; Takkar et al., 2025). Export-oriented supply chains in developing-country contexts responded more to buyer mandates, compliance expectations, and brand-led sustainability programs than to strong domestic regulatory incentives (Akter et al., 2025; Rafid et al., 2024; Ramayanti et al., 2025; Rizvi et al., 2025; Sadurya & Selvaranee, 2025). These differences suggest that implementation pathways vary substantially by region and by the source of coercive or normative pressure.

3.6 Sustainability Implications, Stakeholder Perspectives, and Synthesis

Environmental sustainability outcomes were the most concrete and best documented. Reported benefits included improved carbon-footprint accounting, better batch-level life-cycle assessment, reduced waste, greater visibility

of recycled or circular material flows, and support for product-passport systems (Alves et al., 2024; Carrières et al., 2022; Chowdhury, 2025; Gazzola et al., 2025; Manfrino, 2024; Sadurya & Selvaranee, 2025; Tabassum et al., 2025; Takkar et al., 2025; Wang et al., 2024). Economic outcomes were also relatively visible, including audit efficiencies, reduced documentation burdens, anti-counterfeiting potential, operational savings, and reputational benefits for brands capable of marketing verified provenance (Ahmed & MacCarthy, 2021; Benstead et al., 2024; Carrières et al., 2022; Chowdhury, 2025; Dhillon, 2024; Gazzola et al., 2025; Manfrino, 2024; Sovtić et al., 2025; Tabassum et al., 2025). By contrast, social outcomes were more often asserted than demonstrated. Many papers presented labor transparency, worker-rights visibility, and ethical-compliance monitoring as major reasons to adopt blockchain, yet direct evidence of improved worker conditions or upstream labor practices remained limited (Akter et al., 2025; Alves et al., 2024; Cuc, 2023; Havryliuk, 2025; Hindarto et al., 2024; Mogos & Fragapane, 2022; Rafid et al., 2024; Ramayanti et al., 2025; Rizvi et al., 2025; Sadurya & Selvaranee, 2025).

Stakeholder perspectives were similarly differentiated. Manufacturers and suppliers were primarily concerned with cost, interoperability, workflow disruption, and the burden of digital record generation (Akter et al., 2025; Hindarto et al., 2024; Moretto & Macchion, 2022; Rafid et al., 2024; Ramayanti et al., 2025; Rizvi et al., 2025; Sadurya & Selvaranee, 2025). Brands and retailers emphasized provenance claims, consumer trust, supplier-risk management, and regulatory preparedness (Ahmed & MacCarthy, 2021; Benstead et al., 2024; Cuc, 2023; Gazzola et al., 2025; Manfrino, 2024; Moretto & Macchion, 2022). Consumer-facing studies suggested that buyers valued authenticated information and trustworthy disclosure, but also indicated that transparency tools must be easy to use and embedded in the purchase experience if they are to influence real behavior (Dhillon, 2024; Guo et al., 2023; Sovtić et al., 2025). Across these perspectives, the most convincing pathway was not blockchain alone, but a layered model in which digital capture, permissioned governance, and focal-firm coordination jointly translated ledger functionality into usable transparency. This cross-cutting interpretation is presented schematically in Figure 1.

4. Discussion

The revised synthesis highlights a central disagreement in the literature. Optimistic studies present blockchain as a trust-building infrastructure that can make provenance, audit trails, and sustainability claims more credible. More skeptical studies emphasize that blockchain does not remove cost asymmetries, supplier reluctance, data-quality risks, or buyer power in fragmented apparel chains. This disagreement is itself an important finding: blockchain's value is not determined by ledger design alone, but by implementation conditions, governance arrangements, and the distribution of incentives across tiers.

4.1 How Does Blockchain Enable Transparency

Across the included studies, blockchain enabled transparency through three mechanisms with different levels of evidentiary strength. First, the strongest evidence concerned tamper-resistant audit trails and provenance continuity. Case and pilot studies showed that time-stamped records could support fiber-origin verification, audit readiness, product authentication, and batch-level environmental assessment (Ahmed & MacCarthy, 2021; Carrières et al., 2022; Chowdhury, 2025; Tabassum et al., 2025). Second, moderate evidence supported real-time or near-real-time data sharing when blockchain was combined with IoT, RFID, QR/NFC identifiers, ERP/MIS systems, or digital product-passport infrastructure (Abreu et al., 2025; Akter et al., 2025; Faridi et al., 2023; Karim & Talukder, 2024; Tabassum et al., 2025; Takkar et al., 2025; Wang et al., 2024). These studies suggest that the ledger becomes useful when it is embedded in a broader data-capture stack rather than used in isolation. Third, the weakest evidence concerned accountability for social sustainability. Several papers argued that blockchain could document labor compliance, but few directly measured worker-level outcomes, wage effects, working conditions, or grievance resolution. Thus, the mechanisms are uneven: provenance and environmental-accounting mechanisms are comparatively stronger, while social-accountability mechanisms remain underdeveloped.

4.2 Source-Data Credibility and the Garbage-In Problem

A key limitation of blockchain-based transparency is that ledger immutability does not guarantee truth at the point of data entry. A false record on an immutable ledger may become a permanent and auditable falsehood. This issue is particularly important in fashion supply chains, where upstream records may still be entered manually, supplier declarations may be strategically incomplete, and subcontracting relationships may be hidden. Hindarto et al. (2024) are useful in this respect because their qualitative evidence points to the need for provenance verification beyond the ledger itself. Future systems should therefore combine blockchain with trusted oracles, physical verification, sensor-based capture, third-party certification, and audit protocols at the point where material, labor, or environmental data are first generated.

4.3 Implementation Barriers as Power and Transaction-Cost Problems

The barriers identified in Section 3.4 should not be understood as merely technical. Transaction-cost logic helps explain why upstream suppliers have weak incentives to adopt costly transparency tools when most reputational and regulatory benefits accrue to brands or retailers. Small suppliers must pay for tagging, software integration, staff training, and documentation routines, but they may not receive higher margins or longer contracts in return. At the same time, buyer-driven supply chains can make transparency a compliance requirement rather than a jointly financed governance infrastructure. This creates a systemic barrier: the actors whose data are most needed for credible transparency are often those with the least bargaining power and the weakest capacity to digitize their operations.

4.4 The Missing Social Dimension

The review found a clear imbalance across sustainability dimensions. Environmental and economic outcomes were more commonly measured through carbon accounting, waste tracking, audit time, cost savings, or compliance readiness. Social outcomes were more frequently invoked as a rationale than evaluated as an outcome. This may reflect several factors: blockchain is technically better suited to tracking objects, batches, certificates, and events than lived labor conditions; worker-level data are politically sensitive and difficult to collect; and brand-led transparency programs may prioritize externally reportable product information over worker voice. Future research should measure social outcomes directly, for example by linking blockchain-enabled compliance records with anonymized worker survey data, grievance-channel use, corrective-action completion, training records, wage-payment verification, or occupational-safety indicators.

The maturity gap exists because environmental and economic outcomes can often be represented by product-, batch-, or process-level records, whereas social sustainability depends on worker-level experience, power relations, confidentiality, and enforcement capacity that are harder to capture through ledger data alone.

4.5 Concrete Policy Takeaways

The findings suggest three policy-relevant implications. First, digital product passport schemes should not track only materials, carbon, and circularity attributes. If labor-related fields are absent, such passports may improve product traceability while doing little for labor rights. Minimum labor-related data fields could include verified production site, certification status, training records, grievance-channel availability, and corrective-action status. Second, policymakers should pair transparency mandates with supplier-support mechanisms, including interoperable standards, shared identifiers, subsidized onboarding, and low-burden data-capture tools for small and medium-sized suppliers. Third, regulations should require source-data assurance, not merely ledger storage, because immutable records are valuable only when the underlying input data are credible.

4.6 Implications for Managers and Researchers

For managers, the review suggests that blockchain projects should be scoped around specific transparency problems rather than around the technology itself. Projects aimed at batch traceability, certification integrity, audit readiness, or product-passport reporting are more likely to produce measurable value than broad claims of end-to-end transparency unsupported by data-capture infrastructure. For researchers, the most urgent need is not only more longitudinal research in general, but sharper studies of where transparency breaks down: upstream supplier costs, worker-level data availability, source-data fraud, and comparisons between blockchain and non-blockchain alternatives such as shared audit platforms or interoperable ERP systems.

The resulting evidence map can be summarized as follows: mature evidence clusters around provenance, audit readiness, environmental accounting, and product authentication; intermediate evidence concerns IoT- and product-passport-enabled data sharing; and least-developed evidence concerns worker-level social outcomes, supplier-side incentives, and comparisons with non-blockchain traceability systems. Future research should therefore prioritize longitudinal, multi-tier studies that test whether blockchain adds value beyond interoperable non-blockchain systems and whether transparency gains translate into measurable supplier and worker outcomes.

4.7 Limitations

This review has limitations. First, the review was built from a retained corpus that preserved the screened set and extraction content but not auditable raw database-by-database retrieval counts; this limits the precision with which the identification stage can be reconstructed. Second, the included literature was heterogeneous in design, maturity, and reporting quality, limiting direct comparability. Third, some papers addressed textile or apparel

contexts more broadly rather than fast fashion narrowly defined. The revised synthesis therefore distinguishes direct fast-fashion/apparel evidence from transferable textile evidence. Fourth, because the review was scoped to primary studies and did not undertake formal critical appraisal, the synthesis emphasizes mapping and interpretation rather than weighted causal inference. These limitations define the evidentiary boundary within which the current conclusions should be read.

5. Conclusions

Blockchain has become a prominent technological response to the transparency deficit of fast-fashion supply chains, but the present review shows that primary evidence remains concentrated in prototypes, conceptual system designs, and a relatively small number of pilots and case studies. Across 31 primary studies, blockchain most consistently contributed to traceability and provenance documentation when implemented through permissioned, or invite-only, architectures integrated with IoT, smart contracts, enterprise systems, or product-passport tools. Reported benefits included faster audits, improved batch-level visibility, waste reduction, and stronger sustainability reporting, while persistent barriers included cost, interoperability, fragmented governance, weak upstream digital capacity, and source-data credibility risks. The evidence therefore supports a qualified conclusion: blockchain is neither an unsubstantiated technological promise nor a self-sufficient solution. Its value for fast fashion lies in enabling trustworthy coordination across supply-chain actors, but only when technical design, incentives, governance, and source-data verification are aligned well enough to make transparency operational rather than merely promised.

Taken together, these findings support broader sustainability transitions by showing that digital transparency can connect environmental accounting, economic coordination, and social accountability only when governance standards and policy incentives align the interests of brands, suppliers, regulators, and consumers.

6. Research Gaps and Future Directions

Three gaps remain especially important: (1) direct evidence that blockchain improves labor rights, wages, safety, or worker voice remains minimal; (2) upstream supplier costs and incentives are still usually examined from brand or system-designer perspectives rather than from supplier perspectives; and (3) the incremental value of blockchain over well-governed non-blockchain traceability systems has rarely been tested. These gaps should guide the next generation of empirical research.

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Data Availability

Not applicable.

Conflicts of Interest

The author declares no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

Generative AI and AI-assisted technologies were used only for language editing and document formatting during manuscript preparation. The author reviewed and takes full responsibility for the content of the work.

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Appendix

Country/region and context are reported as represented in the retained review record.

Table A1. Included primary studies and basic characteristics

Study	Country/Region	Study Design or Evidence Type	Research Context	Main Transparency Contribution
Hindarto et al. (2024)	Indonesia	Qualitative/observational inquiry	Fashion industry; manufacturers and supply-chain actors	Provenance-verification needs and source-data credibility challenges.
Moretto & Macchion (2022)	Global/not site-specific in review record	Multi-case/case-based investigation	Fashion supply chains; brands and supply-chain managers	Adoption drivers, barriers, and traceability conditions.
Cuc (2023)	Global	Qualitative/observational sector inquiry	Textile-fashion firms and sector cases	Use cases for traceability, authenticity, and sustainability documentation.
Rizvi et al. (2025)	Pakistan	Survey-based quantitative analysis	Textile sector; industry respondents	Association between blockchain transparency and sustainable supply-chain performance.
Bullón Pérez et al. (2020)	Not site-specific in review record	Case-based/model-development study	Ready-to-wear clothing; supply-chain actors	Product-identity and process-record model for garment traceability.
Ahmed & MacCarthy (2021)	Europe	Case study	Fiber producer and downstream textile-apparel partners	Fiber-origin verification, data integrity, and audit readiness.

Tabassum et al. (2025)	Apparel/textile supply chains	Design-science/prototype-development study	Managers and system users	Faster tracing, reduced audit preparation, and reduced data mismatch.
Guo et al. (2023)	Global/sector-level	Conceptual architecture paper	Sustainable fashion supply chains	Operational transparency and environmental reporting logic without deployment evidence.
Alves et al. (2024)	Portugal/Europe	Design-science/prototype-development study	Textile and clothing value-chain firms	Digital-passport traceability platform for environmental and social indicators.
Carrières et al. (2022)	Australia/New Zealand-China textile chain	Case-based investigation	Wool-textile processing chain	Batch-specific traceability changed life-cycle assessment estimates.
Chen et al. (2021)	Brand-clothing context	Design-science/prototype-development study	Brands and consumers	Hyperledger authentication and traceable anti-counterfeit management.
Faridi et al. (2023)	Apparel Industry 4.0	Design-science/prototype-development study	Manufacturers, retailers, and logistics actors	Blockchain-IoT-RFID event-level traceability framework.
Dhillon (2024)	Global fashion sector	Survey-based quantitative analysis	Industry respondents	Perceived transparency, traceability, and sustainability benefits.
Karim & Talukder (2024)	Transparency-critical supply chains including apparel export chains	Survey-based quantitative analysis	Supply-chain respondents	Integrated IoT-blockchain capability showed strongest association with real-time transparency.
Benstead et al. (2024)	Sweden, Italy, Germany	Multi-case/case-based investigation	Entrepreneurs, start-ups, and brands	Blockchain use for provenance exposure and consumer-facing transparency.
Mogos & Fragapane (2022)	Europe	Mixed-method study	Circular and transparent value-chain project stakeholders	Circular value-chain documentation and stakeholder visibility.
Pal & Yasar (2021)	Textile/apparel SCM context	Conceptual architecture paper	Supply-chain actors	IoT-blockchain event-visibility and logistics-record functions.
Sadurya & Selvaranee (2025)	India	Mixed-method/case-based model	Textile sector; firm managers and chain actors	Tracking of waste, energy, and sustainable-practice documentation.
Sovtić et al. (2025)	Fashion retail market	Survey-based quantitative analysis	Consumers	Readiness to buy blockchain-authenticated fashion products.
Akter et al. (2025)	Bangladesh	Mixed-method study	Fashion value chain; firms and key informants	Zero-waste practice tracking and implementation barriers.
Manfrino (2024)	Europe	Multi-case/case-based investigation	Brands and compliance actors	Product-passport and CSRD-oriented sustainability documentation.
Abreu et al. (2025)	Europe	Design-science/prototype-development study	Value-chain stakeholders	Design principles for blockchain-based digital product passports.
Wang et al. (2024)	Cotton-textile chain	Design-science/prototype-development study	Supply-chain nodes	Token-linked cotton-batch traceability.
Takkar et al. (2025)	Complex textile commodity chains	Design-science/prototype-development study	Supply-chain actors	Traceability extended to forward and return flows for circular-economy transparency.
Gazzola et al. (2025)	Fashion companies	Multi-case/case-based investigation	Three fashion companies; firms and managers	Technology-supported circular fashion practices and transparency-oriented sustainability data.
Chowdhury (2025)	United States-linked apparel sourcing	Case-based investigation	Brand and suppliers	Waste, cost, and carbon-footprint reductions in a bounded case.
Majumdar (2025)	Apparel industry	Conceptual architecture paper	Enterprise users	Platform-level data-sharing proposal.
Ramayanti et al. (2025)	Indonesia	Qualitative/observational inquiry	Local creative batik and weaving producers	Authentication and product traceability for local creative products.
Havryliuk (2025)	Global/business processes	Conceptual architecture paper	Firms and marketers	Provenance records connected with sustainability-oriented business processes.

Shakir (2025)	Wool-textile chain	Conceptual architecture paper	Producers and processors	Farm-to-fabric monitoring for wool provenance and process documentation.
Rafid et al. (2024)	Bangladesh and China	Multi-case/case-based investigation	Garments industry cases	Adjacent evidence on digital transparency and supply-chain resilience.

Note: Study rows follow their original order of first mention. PLS-SEM, partial least squares structural equation modeling; UTAUT2, Extended Unified Theory of Acceptance and Use of Technology; LCA, life cycle assessment; IoT, Internet of Things; ERP, enterprise resource planning; MIS, management information system; SCM, supply-chain management; NFT, non-fungible token; CSRD, Corporate Sustainability Reporting Directive.